

Quiz

Name : _____

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Example 3.20 Suppose a binary signal $S \in -1, 1$ with $P(S = 1) = p$ is transmitted, but the signal is received with standard normally distributed noise N , so that we only observe $Y = N + S$. Bayes Rule says the probability that $S = 1$ given an observed value y is

$$P(S = 1|Y = y) = \frac{P(S = 1)f_{Y|S}(y|1)}{f_Y(y)} = \frac{\frac{p}{\sqrt{2\pi}}e^{-(y-1)^2/2}}{\frac{p}{\sqrt{2\pi}}e^{-(y-1)^2/2} + \frac{1-p}{\sqrt{2\pi}}e^{-(y+1)^2/2}} = \frac{pe^y}{pe^y + (1-p)e^{-y}}$$

- (a) What happens to the probability that the $S=1$ if y becomes very negative?
- (b) What is the probability that the signal is 1 if $y = 0$?
- (c) Why was it valid that $f_Y(y) = \frac{p}{\sqrt{2\pi}}e^{-(y-1)^2/2} + \frac{1-p}{\sqrt{2\pi}}e^{-(y+1)^2/2}$ in the denominator?

1. True or False?

- (a) Given two random variables X and Y , the joint PDF must be such that

$$\int_{-\infty}^{\infty} f_{X,Y}(x, y)dx = 1$$

- (b) A valid pdf must take on finite values.
- (c) The expectation of a product of 2 random variables is the product of their expectations.
- (d) The expectation of a sum of 2 random variables is the sum of their expectations.
- (e) If X and Y are independent, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$